* Let there be $n=14$ jobs for sequencing whose profits are 36, 21, 28, 19, 21, 13, 28, 25, 18, 20, 27, 22, 14, 26. Times are 5, 8, 3, 4, 4, 9, 12, 14, 2, 7, 5, 1, 6, 3.

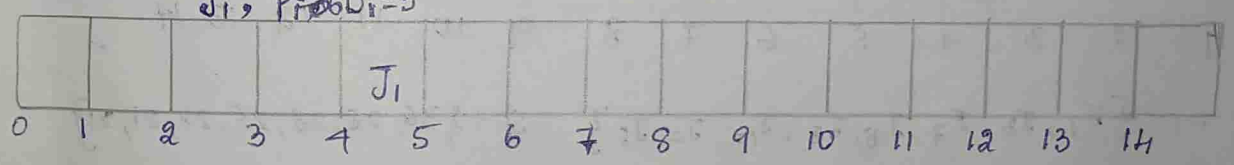
Step-1

	Profit	Time	Pro/Time
J_1	36	5	7.2
J_3	28	3	9.33
J_7	28	12	2.3
J_{11}	27	5	5.4
J_{14}	26	3	8.6
J_8	25	14	1.7
J_{12}	22	1	22
J_2	21	8	2.6
J_5	21	4	5.2
J_{10}	20	7	2.86
J_4	19	4	4.75
J_9	18	2	9
J_{13}	14	6	2.33
J_6	13	9	1.44

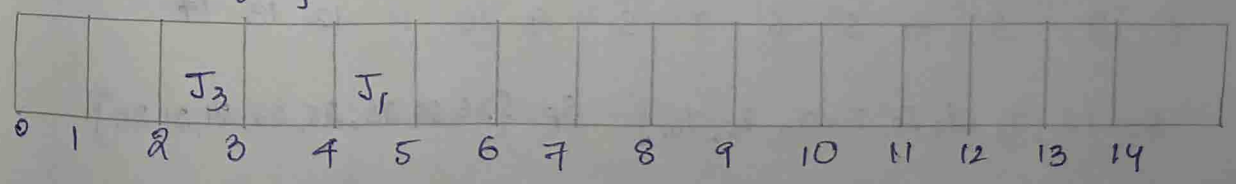
Given,
No. of jobs (N) = 14

Step-2 :- Create Slots and Assign jobs

J_1, $P_1=36$, $D_1=5$



$P=\{36\}$
$J=\{J_1\}$



$S=\{J_1, J_3\}$ $S_P=\{36, 28\}$

		J_3		J_1		J_7				J_7			

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7\}$ $\quad S_p = \{36, 28, 28\}$

		J_3		J_1						J_7			

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_{11}\}$ $\quad S_p = \{36, 28, 28, \}$

		J_3		J_1							J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, \cancel{J_{11}}, J_8\}$ $\quad S_p = \{36, 28, 28, \cancel{24}, 25\}$

J_{12}		J_3		J_1							J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_{11}, J_8, J_{12}\}$ $\quad S_p = \{36, 28, 28, \cancel{24}, 25, 22\}$

J_{12}		J_3		J_1			J_2				J_7	J_8	

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_8, J_{12}, J_2\}$ $\quad S_p = \{36, 28, 28, 25, 22, 21\}$

J_{12}		J_3	J_5	J_1			J_2				J_7	J_8	

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S_1 = \{J_1, J_3, J_7, J_8, J_8, J_{12}, J_2, J_5\}$ $\quad S_p = \{36, 28, 28, 25, 22, 21, 21\}$

J_{12}		J_3	J_5	J_1		J_{10}	J_2				J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_8, J_{12}, J_2, J_5, J_{10}\}$ $\quad S_p = \{36, 28, 28, 25, 22, 21, 21, 20\}$

J_{12}	J_9	J_3	J_5	J_1		J_{10}	J_2				J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_8, J_{12}, J_2, J_5, J_{10}, J_9\}$ $S_p = \{36, 28, 28, 25, 22, 21, 21, 20, 18\}$

J_2	J_9	J_3	J_5	J_1	J_{13}	J_{10}	J_2				J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_8, J_{12}, J_2, J_5, J_{10}, J_9, J_{13}\}$ $S_p = \{36, 28, 28, 25, 22, 21, 21, 20, 18, 14$

J_{12}	J_9	J_3	J_5	J_1	J_{13}	J_{10}	J_2	J_6			J_7		J_8

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

$S = \{J_1, J_3, J_7, J_8, J_{12}, J_2, J_5, J_{10}, J_9, J_{13}, J_6\}$

$S_p = \{36, 28, 28, 25, 22, 21, 21, 20, 18, 14, 13\}$.

② Using Haffmann coding, arrive max encoding bits for pharase;

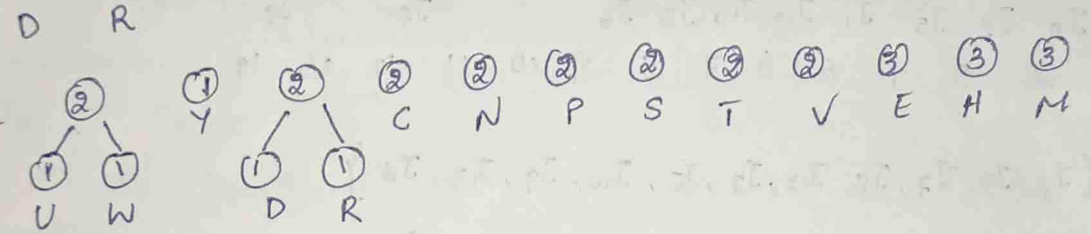
AMRITA VISWAVIDYAPEETHAM CHENNAI CAMPUS.

A	M	R	I	T	V	S	H	W	D	Y	P	E	C	N	U
7	3	1	4	2	2	2	3	1	1	1	2	3	2	2	1

Arrange in ascending order:

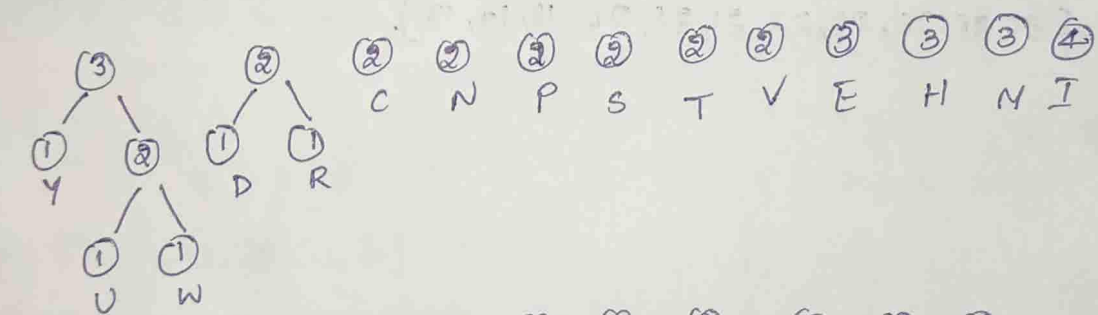
1	1	1	1	1	2	2	2	2	2	2	3	3	3	3	4
D	R	U	W	Y	C	N	P	S	T	V	E	H	M		I

S_1

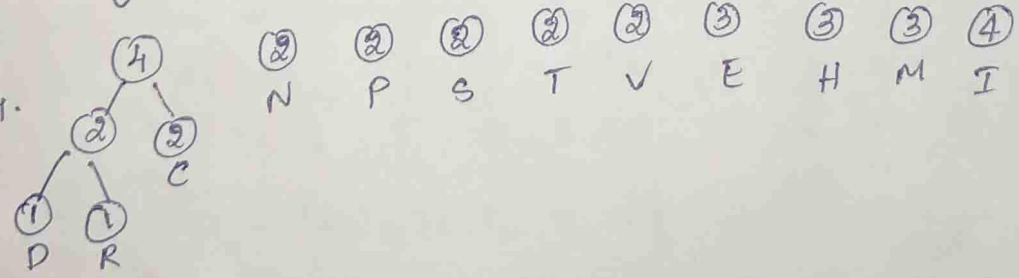


S_2

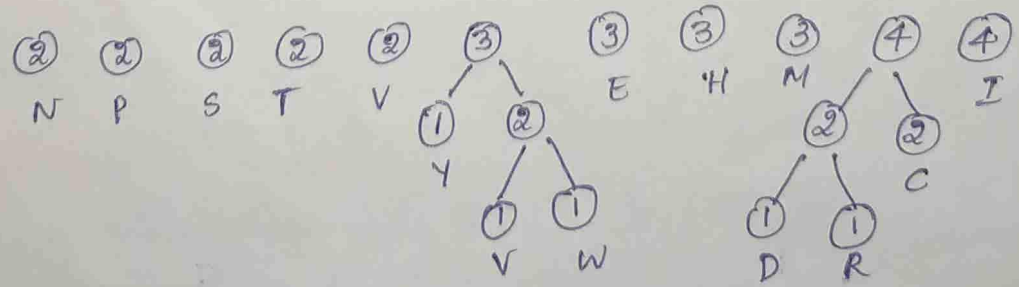
S_3



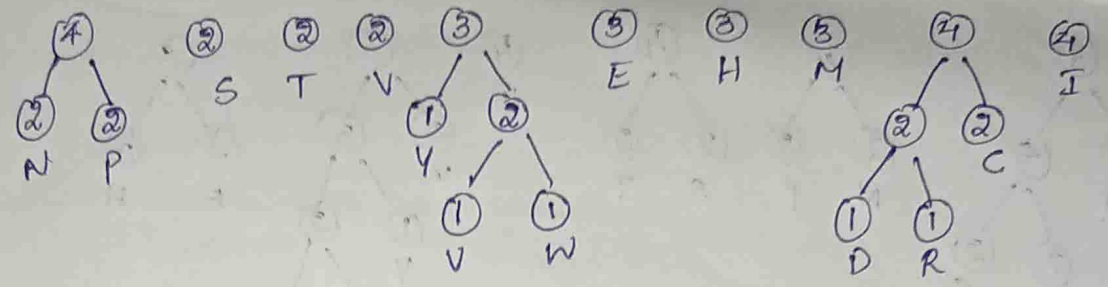
S_4.



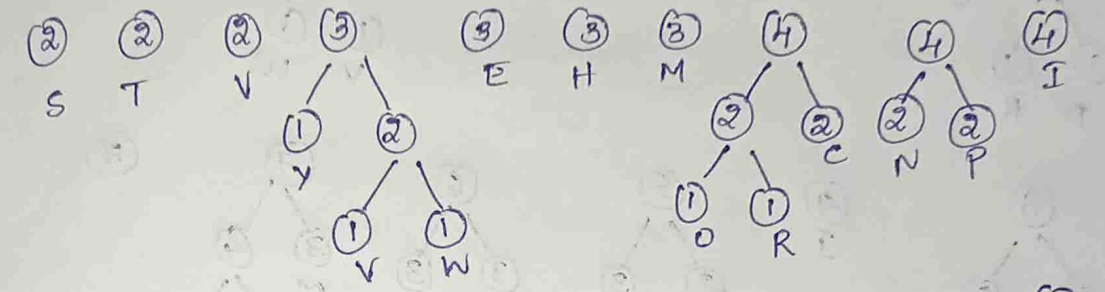
S_5



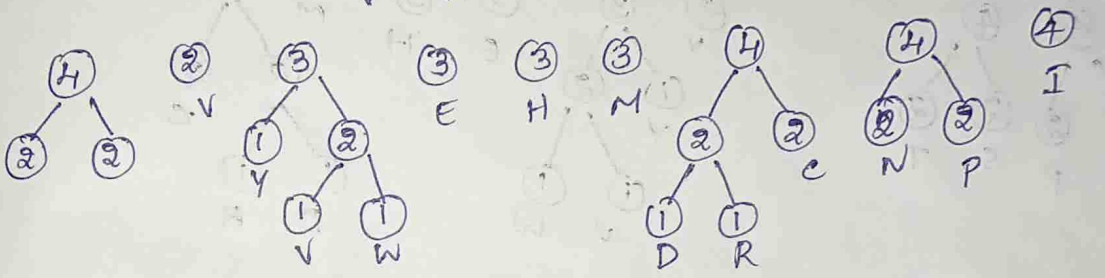
S6



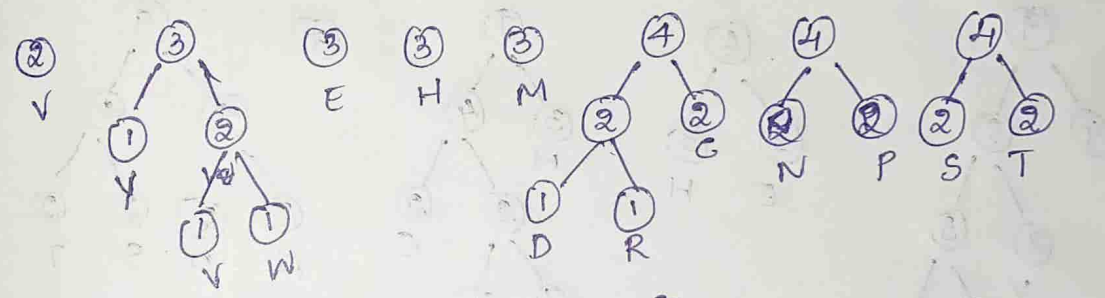
S7



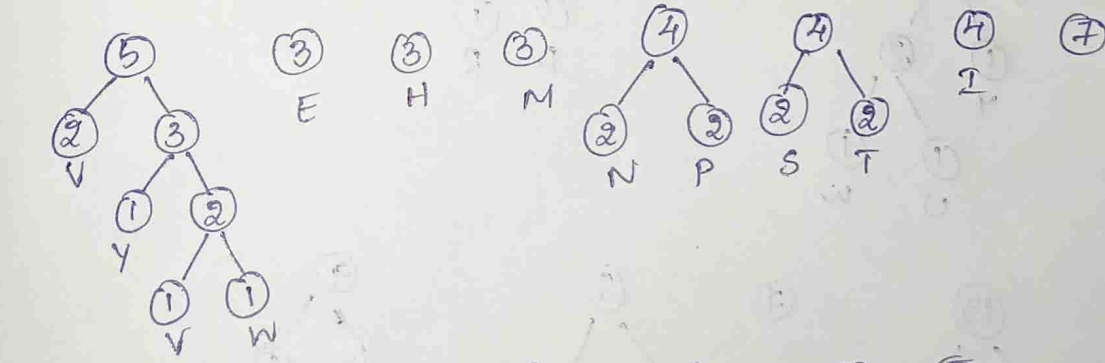
S8



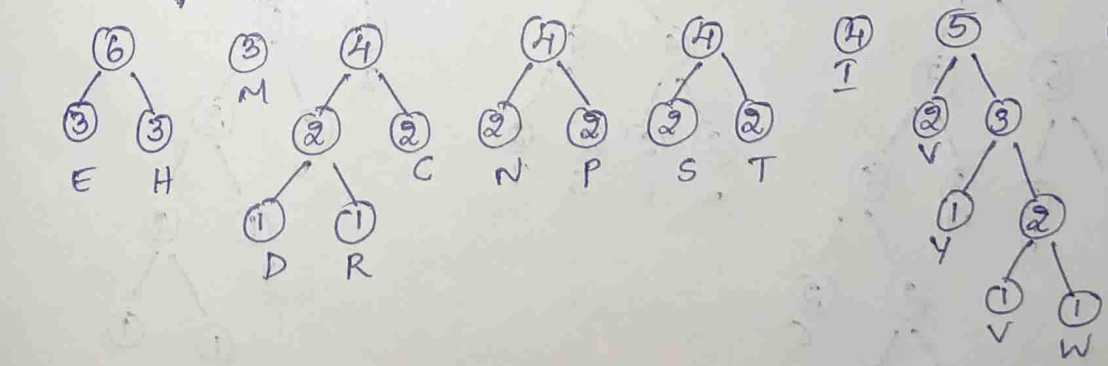
S9.



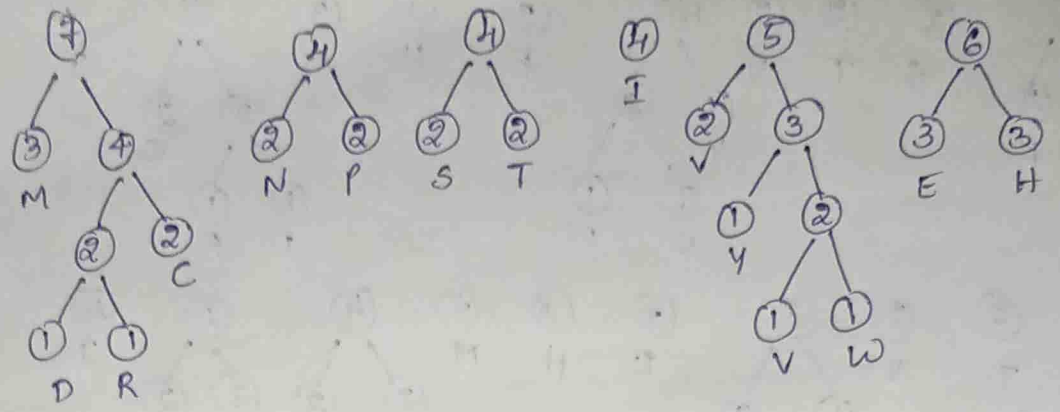
S10



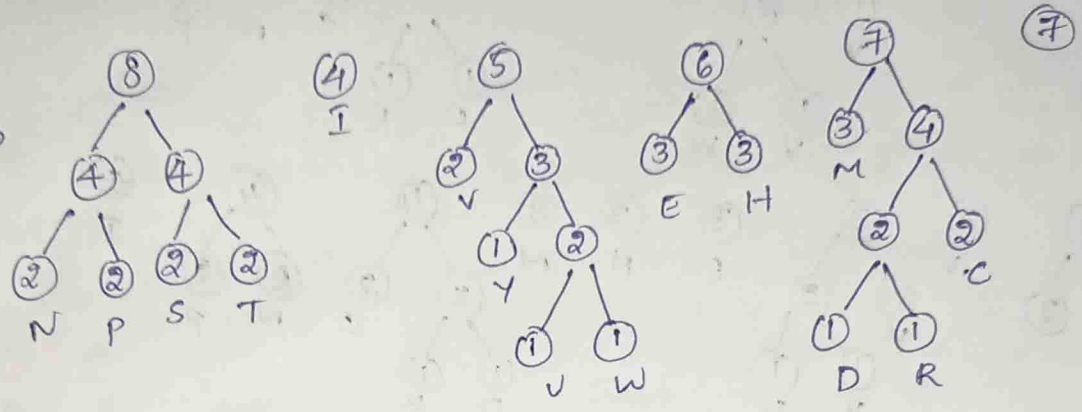
S11



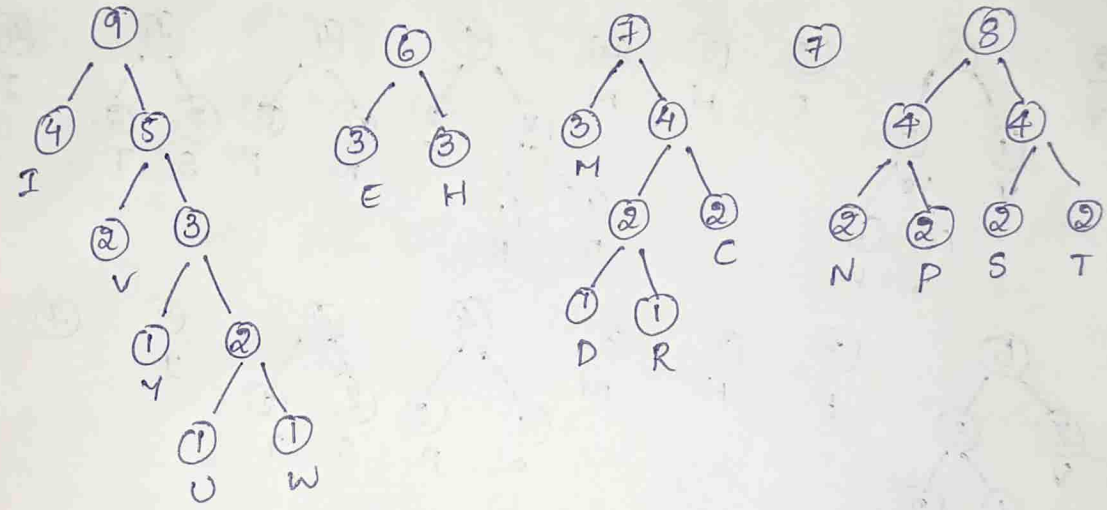
S_{12}



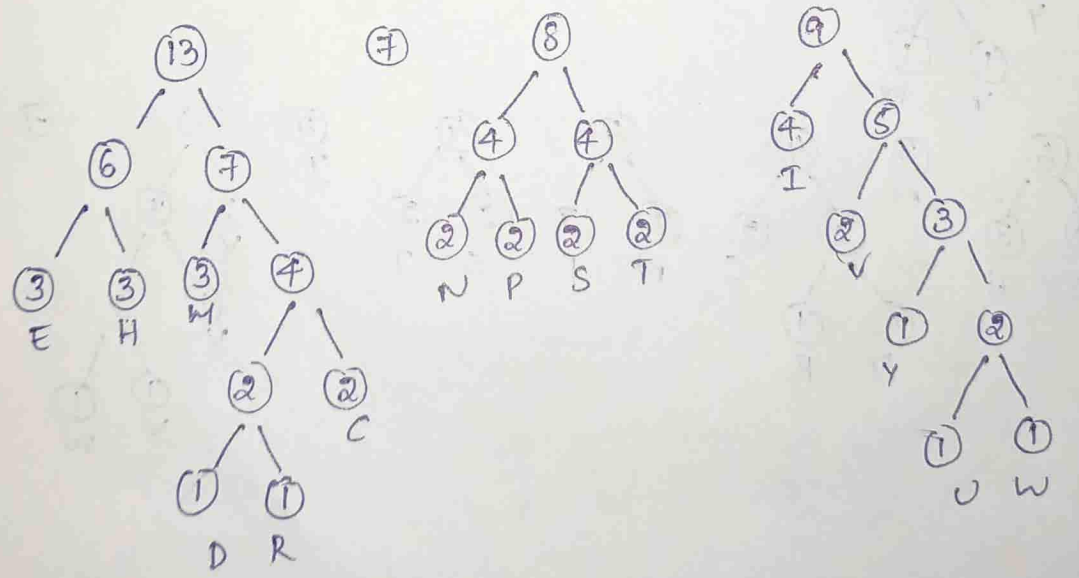
S_{13}

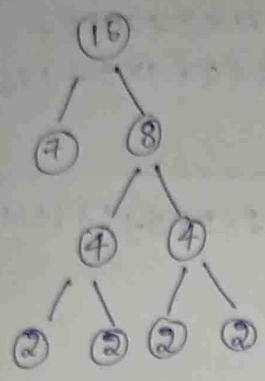


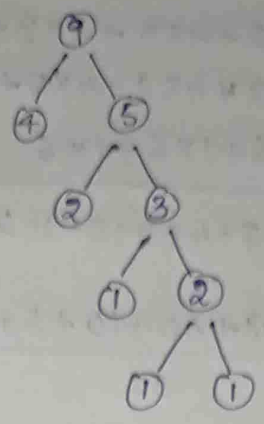
S_{14}

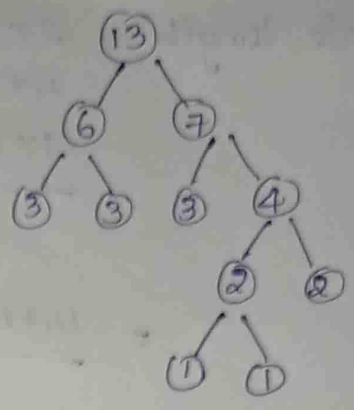


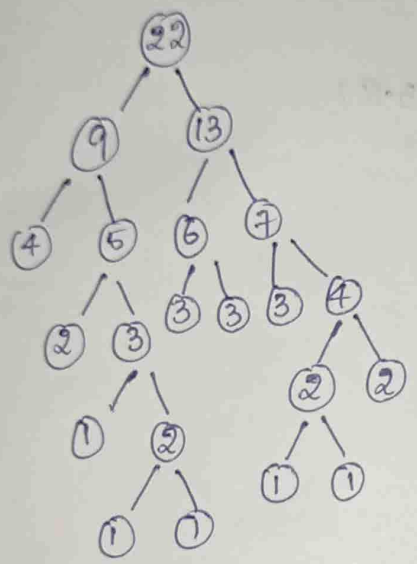
S_{15}

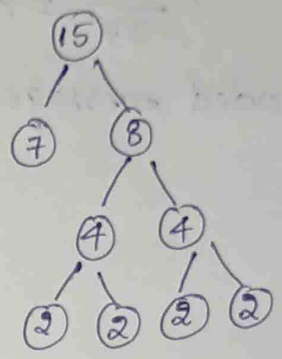


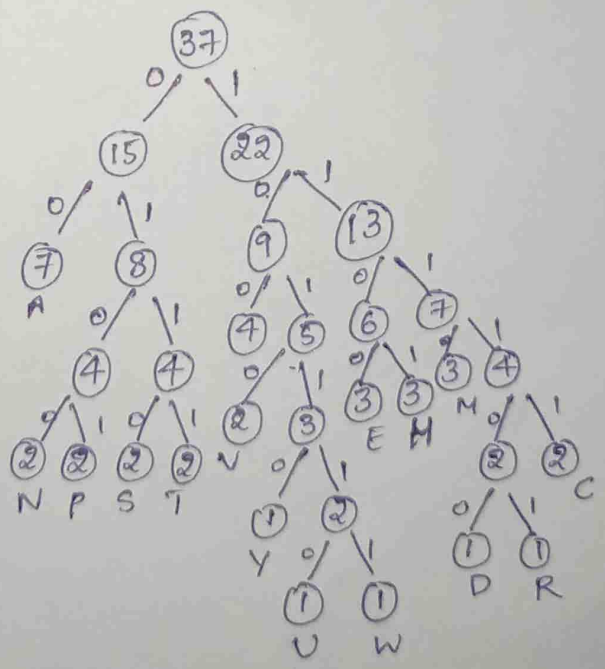












A - 00

I - 100

V - 1010

E - 1100

H - 1101

M - 1110

N - 0100

P - 0101

S - 0110

T - 0111

Y - 10110

U - 101110

W - 101111

D - 111100

R - 111101

C - 11111

$$\text{Avg code length} = \frac{7\times2+4\times3+4\times3+4\times3+4\times3+4\times2+4\times2+4\times2+4\times2+4\times2+5\times2+1\times5+1\times6+1\times6+1\times6+1\times6}{7+4+3+3+3+2+2+2+2+2+2+1+1+1+1+1}$$

$$= \frac{14+48+40+10+5+24}{37}$$

$$= \frac{141}{37} = 3.81$$

Length of encoded message $\Rightarrow 37\times3.81$

$\Rightarrow 141.$